Alice Paul

Contact Information

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Academic Appointments

Associate Professor of Biostatistics, Teaching Scholar
Brown University, 2025–Present.

Director, Undergraduate Concentration in Statistics
Brown University, 2024–Present.

Associate Director, Master's Program in Biostatistics
Brown University, 2023–Present.

Assistant Professor of Biostatistics, Teaching Scholar
Brown University, 2020–2025.

Assistant Professor of Applied Mathematics and Computer Science
Franklin W. Olin College of Engineering, 2019–2020.

Postdoctoral Research Associate
Department of Biostatistics and the Data Science Initiative, Brown University, 2017–2019.
Advised By: Pedro Felzenszwalb.

Education

Cornell University, Ithaca, NY.
Ph.D. Operations Research and Information Engineering, August 2017.
Advised By: David P. Williamson.

Harvey Mudd College, Claremont, CA.
B.S. Mathematics with High Distinction, May 2012.

Honors and Awards

- Young Investigator Award, ASA Teaching Statistics in the Health Sciences (TSHS) Section, 2024.

- Dean's Award for Excellence in Classroom Teaching in the School of Public Health, Spring 2022.
- NDSEG Fellow.

Students advised by me marked by * in publications below. Alphabetical order marked by †

Peer-Reviewed Articles

1. Gaelen P. Adam*, Jay DeYoung, **Alice Paul**, Ian J. Saldanha, MBBS, Ethan M. Balk, Thomas A. Trikalinos, Byron C. Wallace, Literature Search Sandbox: a Large Language Model that generates search queries for systematic reviews, JAMIA Open, 2024 (forthcoming).

2. **Alice Paul** and Susan Martonosi. *The All-Pairs Vitality-Maximization (VIMAX) Problem*, Annals of Operations Research, 2024. https://doi.org/10.1007/s10479-024-06022-4.

3. Pedro Felzenszwalb, Caroline Klivans, and **Alice Paul**†. *Clustering with Semidefinite Programming and Fixed Point Iteration*, Journal of Machine Learning Research, 2022. http://jmlr.org/papers/v23/21-0402.html.

4. Pedro Felzenszwalb, Caroline Klivans, and **Alice Paul**†. *Iterated Linear Optimization*, Quaterly of Applied Mathematics, 2021. DOI: DOI: https://doi.org/10.1090/qam/1594.

5. **Alice Paul** and David Williamson. *Easy Capacitated Facility Location Problems, with Connections to Lot-Sizing*, Operations Research Letters, 2020. DOI: https://doi.org/10.1016/j.orl.2019.12.006.

6. **Alice Paul**, Daniel Freund, Aaron Ferber, David Shmoys, and David Williamson. *Budgeted Prize-Collecting Traveling Salesman and Minimum Spanning Tree Problems*, Mathematics of Operations Research, 2019. DOI: https://doi.org/10.1287/moor.2019.1002.

7. Amariah Becker and **Alice Paul**. *A Framework for Vehicle Routing Approximation Algorithms in Trees*, Algorithms and Data Structures Symposium, 2019.

8. Jacob Feldman, **Alice Paul**†, and Huseyin Topaloglu. *Technical Note: Assortment Optimization with Small Consideration Sets,* Operations Research, 2019. DOI: https://doi.org/10.1287/opre.2018.1803.

9. Jacob Feldman and **Alice Paul**. *Relating the Approximability of the Fixed Cost and Space Constrained Assortment Problems*, Production and Operations Management, 2018. DOI: https://doi.org/10.1111/poms.12983

10. **Alice Paul**, Daniel Freund, Aaron Ferber, David Shmoys, and David Williamson. *Prize-Collecting Traveling Salesman with a Budget Constraint*, European Symposium on Algorithms, 2017.

11. **Alice Paul**, Jacob Feldman, and James Mario Davis. *Assortment Optimization and Pricing under a Nonparametric Tree Choice Model*, Manufacturing and Service Operations Management, 2017. DOI: https://doi.org/10.1287/msom.2017.0662.

12. **Alice Paul**, Matthias Poloczek, and David P. Williamson. *Simple Approximation Algorithms for Balanced MAX 2SAT*, Algorithmica, 2017. DOI: https://doi.org/10.1007/s00453-017-0312-6.

13. **Alice Paul**, Matthias Poloczek, and David P. Williamson. *Simple Approximation Algorithms for Balanced MAX 2SAT*, Latin American Theoretical Informatics Symposium, 2016.

14. **Alice Paul** and Nicholas Pippenger. *A Census of Vertices by Generations in Regular Tessellations of the Plane*, Electronic Journal of Combinatorics, 2011. DOI: https://doi.org/10.37236/574.

Books

1. **Alice Paul**. *Mastering Health Data Science Using R*, CRC Press, 2025. ISBN: 9781032729930.

2. **Alice Paul**. *Mastering Health Data Science Using R*, Online 2023. https://alicepaul.github.io/health-data-science-using-r/

Book Chapters

1. **Alice Paul** and Susan Martonosi. *Operations Research*, in Nathan Carter (ed.), *Data Science for Mathematicians*, 2020. DOI: https://doi.org/10.1201/9780429398292.

2. Daniel Freund, Ashkan Norouzi-Fard, **Alice Paul**, Shane Henderson and David B. Shmoys. *Data-Driven Rebalancing Methods for Bike-Share Systems*, in E. Chrisotomi et al. (ed.), *Analytics for the Sharing Economy: Mathematics, Engineering, and Business Perspectives*, 2020. DOI: https://doi.org/10.1007/978-3-030-35032-1

Software

1. Hannah Eglinton* and **Alice Paul**. R Package: riskscores, 2024.

2. **Alice Paul**, Hannah Eglinton*, Jialin Liu*, Joanna Walsh*, and Xinbei Yu*. R Package: R: HDSinRdata, 2023.

Preprints

1. Hannah Eglinton*, Silvia Chiang, Tongtong Zhao*, Yu Yan*, and **Alice Paul**. *A Computationally Efficient Algorithm for Producing Risk Score Models with Applications to Tuberculosis Diagnosis and Treatment Adherence*, In Submission, 2024.

Panels

1. "Pedagogical Design and Innovations for OR and Supply Chain Education," Committee on Teaching and Learning Teaching Excellence and Networking Workshop, INFORMS, October 2025.

Invited Presentations

2. "Algorithms in Data Science," Applied Math in Statistics and Data Science Education, ICERM, Brown University, May 2025.
3. "Prescriptive Analytics in the Data Analysis Pipeline," Data-Driven Innovations in OR Education, INFORMS, 2023.
4. "Iterative Algorithms for Semidefinite Programming," American Mathematical Society Eastern Sectional Meeting, 2021.
5. "Prize-Collecting TSP with a Budget Constraint," International Symposium on Math Programming, 2018.
6. "Data-Driven Optimization for Bike-Share Systems," Data Science Initiative Colloquium, Brown University, 2017.
7. "Prize-Collecting TSP with a Budget Constraint," European Symposium on Algorithms, 2017.
8. "Assortment Optimization for Choosy Customers," INFORMS, 2016.
9. "Assortment Optimization for Choosy Customers," INFORMS Revenue Management and Pricing Conference, 2016.
10. "Simple Approximation Algorithms for Balanced MAX 2SAT," LATIN, 2016.
11. "Revenue Management under a Nonparametric Ranking-Based Choice Model," INFORMS, 2015.
12. "Detecting Covert Members of Terrorist Networks," Young Women in Discrete Math, 2013.
13. "Detecting Covert Members of Terrorist Networks," INFORMS, 2012.

Grants

1. Brown Data Science Institute Seed Grant (06/2024-08/2025)
   Building a Prescriptive Analytics Research Collaboration with the Rhode Island Public Transit Authority (RIPTA). Role: PI.

2. Brown Data Science Institute Seed Grant (06/2023-08/2024)
   Creating an Online R Resource with a Focus on Public Health. Role: PI.

Teaching Experience

- PHP 2511: Applied Regression Analysis, SP 2026.
- DATA 2020: Statistical Learning, Brown University, SP 2018, SP 2019, SP 2022, SP 2026.
- PHP 1560: Using R for Data Analysis, Brown University, FA 2025.
- PHP 1511: Applied Regression Analysis, Brown University, SP 2024, SP 2025.
- PHP 2690: Optimization Algorithms for Data Science, Brown University, SP 2025.
- PHP 2550: Practical Data Analysis, Brown University, FA 2020, FA 2021, FA 2022, FA 2023, FA 2024.
- PHP 2650: Statistical Learning and Big Data, Brown University, SP 2021, SP 2022, SP 2023, SP 2024.
- PHP 1511/2511: Applied Regression Analysis, Brown University, SP 2023.
- PHP 1560/2560: Statistical Programming in R, Brown University, FA 2020, FA 2021, FA 2022, FA 2023.
- ENGR 3599: Data Structures and Algorithms, Olin College, SP 2020.
- MTH 1111/SCI 1111: Modeling and Simulation of the Physical World, Olin College, FA 2019.
- ENGRI 1101: Engineering Applications of Operations Research, Cornell University, FA 2016.
- ORIE 3310: Optimization II, Cornell University, SU 2015.

Professional Workshops

- Organizing Committee, Applied Math in Statistics and Data Science Education, ICERM, Brown University, May 2025.

Student Workshops (Designed and Led)

- Building a Personal Portfolio Website, Brown University, FA 2022, SP 2024.
- Intro to Python for R Users, Brown University, SP 2023, SP 2024.
- Intro to R for Health Data Science, Brown University, SU 2023.

Advising and Mentoring

Masters Thesis or Capstone Advisor

- Jingrui Liu, Biostatistics Masters Project, Brown University, Fall 2025-Spring 2026.
- Xurui Gao, Biostatistics Masters Thesis, Brown University, Fall 2025-Spring 2026.
- Thomas Arnold, Biostatistics Masters Thesis, Brown University, Fall 2024-Spring 2025.
- William Qian, Biostatistics Masters Thesis, Brown University, Fall 2024-Spring 2025.
- Yingqiu Huang, Biostatistics Masters Thesis, Brown University, Fall 2024-Spring 2025.
- Hannah Eglinton, Biostatistics Masters Thesis, Brown University, Fall 2023-Spring 2024. **Biostatistics Masters Thesis Award**.
- Yu Yan, Biostatistics Masters Thesis, Brown University, Fall 2023-Spring 2024.
- Kyla Finlayson, Biostatistics Masters Project, Brown University, Fall 2022-Spring 2023.
- Antonella Basso, Biostatistics Masters Thesis, Brown University, Fall 2022-Spring 2023.
- Tongtong Zhao, Data Science Masters Capstone, Brown University, Summer 2022.
- Zhirui Li, Data Science Masters Capstone, Brown University, Summer 2022.
- Tobias Dekara, Biostatistics Masters Thesis, Brown University, Fall 2021-Spring 2022.
- Emily Jaekle, Data Science Masters Capstone Advisor, Brown University, Summer 2018.
- Sibel Kadioglu, Data Science Masters Capstone Advisor, Brown University, Summer 2018.

Undergraduate Thesis or Capstone Advisor

- Simone Donoho, Undergraduate Honors Thesis in Applied Math, Brown University, Fall 2025-Spring 2026.
- Seehanah Tang, Undergraduate Honors Thesis in Applied Math, Brown University, Fall 2024-Spring 2025.
- Rayan Ouaddi, Undergraduate Capstone in Statistics, Brown University, Spring 2025.
- Benjamin Moshes, Undergraduate Capstone in Statistics, Brown University, Fall 2024.
- Nicolas Fernandez Baigun, Undergraduate Capstone in Statistics, Brown University, Fall 2024.
- Jania Vandevoorde, Undergraduate Capstone in Statistics, Brown University, Fall 2024.
- Kameel Dossal, Undergraduate Honors Thesis in Statistics, Brown University, Fall 2023-Spring 2024.
- Shreyas Mishra, Undergraduate Honors Thesis in Computer Science, Brown University, Fall 2023-Spring 2024.
- John Chung, Undergraduate Capstone in Statistics, Brown University, Spring 2024.
- Morgan Cunningham, Undergraduate Capstone in Statistics, Brown University, Fall 2023.
- Kyran Flynn, Undergraduate Honors Thesis in Applied Math and Computer Science, Brown University, Spring 2022-Spring 2023. **Finalist of the INFORMS Undergraudate Research Prize 2023.**

PhD Committee Member

- Teressa Chambers, Department of Applied Math, Brown University, Graduated 2025.

Masters Thesis Reader

- Weijing Yin, Biostatistics Masters Thesis, Brown University, Spring 2026.
- Daven Crossland, Public Health Masters Thesis, Brown University, Spring 2024.
- Tova Ibottson, Biostatistics Masters Thesis, Brown University, Spring 2024.
- Yuliia Sereda, Public Health Masters Thesis, Brown University, Spring 2022.

Undergraduate Thesis Reader

- Benjamin Buka, Undergraduate Honors Thesis in International and Public Affairs, Brown University, Spring 2026.

- Heon Lee, Undergraduate Honors Thesis in Applied Math and Computer Science, Brown University, Spring 2026.
- Max Collins, Undergraduate Thesis in Mathematics, Harvey Mudd College, Spring 2025.
- Akash Singirikonda, Undergraduate Honors Thesis in Computer Science, Brown University, Spring 2025.
- Kenyon Graham, Undergraduate Honors Thesis in Biology, Brown University, Spring 2025.

Independent Study Supervisor

- Jacy Weems, PhD Independent Study on LLMs for Qualitative Research, Brown University, Spring 2025–Present.
- Gaelen Adam, PhD Independent Study on Sequence Models, Brown University, Fall 2022.
- Victoria McDermott, Independent Study, Olin College, Fall 2019.
- Pravallika Dhulipalla, Independent Study, Olin College, Fall 2019.

Other Research Advising

- Simone Donoho, Undergraduate Research Student, Brown University, Summer 2025.
- Shengning (Leo) Zhang, Undergraduate Research Student, Brown University, Summer 2025.
- Michael Kratzer, Undergraduate Research Student, Brown University, Summer 2025.
- Benjamin Moshes, Undergraduate Research Student, Brown University, Spring 2025.
- Kyran Flynn, Undergraduate Research Student, Brown University, Spring 2022.
- Cassandra Overney, Undergraduate Research Student, Olin College, Fall 2019-Spring 2021.
- Manu Patil, Undergraduate Research Student, Olin College, Spring 2020.
- Daniel Suh, Undergraduate Research Student, Brown University, Summer 2018.

Department Service

- School of Public Health Associate Dean of Education Search Committee, Spring 2026 – Summer 2026.

- Chair of the Masters in Biostatistics Admission Committee. Department of Biostatistics, Brown University, Spring 2021-Current.
- Data Science Advisory Board, Fall 2021-Current.
- Online MPH Faculty Advisory Committee, Fall 2021 - Current.
- Biostatistics Academic Program Committee, Fall 2021 - Current.
- Masters of Public Health Core Adviser, Brown University, Fall 2021-Current.
- Biostatistics Introductory Course Committee, Fall 2022-Spring 2023.
- Online MPH Teaching Scholar Search, Fall 2021-Spring 2022.
- Biostatistics Teaching Scholar Search Committee, Fall 2021-Spring 2022.
- Nora Kahn Piore Award Selection Committee. School of Public Health, Brown University, Spring 2022.
- Masters of Public Health Admissions Committee. School of Public Health, Brown University, Spring 2021.

Professional Service

- Board of Directors, Biostats4You, University of Minnesota's Clinical and Translational Science Institute, Fall 2024-Current.
- External Advisor Board, NSF Artificial Intelligence Research Institute for Advances in Optimization (AI4OPT), Spring 2024-Current.
- Judge, ASA Data Visualization Poster Competition for Grades K–12, Spring 2024.

Teaching Training

- Sheridan Center Problem-Solving Course Design Institute, Spring 2023.
- Sheridan Center Junior Faculty Teaching Fellows Program. Brown University, Fall 2020 - Spring 2021.
- Sheridan Center Anchor Program. Brown University, Fall 2020.

Editorial Board Activities

- INFORMS Transactions on Education Editorial Board, Fall 2022-Present.

Reviewer

INFORMS Transactions on Education, INFORMS Journal on Computing, Harvard Data Science Review, Mathematical Programming, Production and Operations Management, Operations Research, Algorithmica, Operations Research Letters, SIAM Journal of Discrete Mathematics, Probability in the Engineering and Informational Sciences.

Professional Societies

Institute for Operations Research and Management Sciences (INFORMS), American Statistical Association (ASA), ASA Section on Teaching of Statistics in the Health Sciences (TSHS).